

Customer Churn Analysis

Project Overview

Customer churn is an important business problem for subscription-based companies. This project analyzes telecom customer data to identify patterns associated with customer churn and understand which customer segments are more likely to leave.

The project uses Python for data preparation, SQL for business analysis, and Tableau for visualization.

Objective

The main objectives of this project are to:

- Calculate the overall customer churn rate.
- Identify customer segments with higher churn.
- Analyze churn based on contract type, tenure, payment method, and internet service.
- Compare monthly charges between churned and retained customers.
- Analyze the relationship between technical support and churn.
- Present the findings through an interactive Tableau dashboard.

Dataset

The Telco Customer Churn dataset contains information about 7,043 customers.

It includes customer details such as:

- Demographics
- Tenure
- Contract type
- Internet service
- Payment method
- Technical support
- Monthly charges
- Total charges
- Churn status

The cleaned dataset was used for the final SQL and Tableau analysis.

Tools & Technologies

- Python - Data cleaning and preprocessing
- Pandas - Data manipulation and analysis
- MySQL - SQL-based business analysis
- Tableau - Dashboard and visualization
- Git & GitHub - Version control and project documentation

Data Preparation

The dataset was initially inspected using Python to understand its structure, data types, and data quality.

During preprocessing:

- Missing and blank values were checked.
- Data types were reviewed and corrected where necessary.
- TotalCharges was converted into a numeric field.
- A TenureGroup feature was created to categorize customers based on their tenure.
- The cleaned dataset was exported for further analysis.

The final cleaned dataset contained 7,043 records.

Analysis Performed

SQL was used to analyze customer churn from different business perspectives.

The analysis covered:

- Overall churn rate
- Churn by contract type
- Churn by tenure group
- Churn by payment method
- Churn by internet service
- Monthly charges by churn status
- Churn by senior citizen status
- Churn by technical support

These analyses were used to identify customer groups that may require greater attention from the business.

Tableau Dashboard

A Tableau dashboard was created to provide a visual summary of the analysis.

Key KPIs

- Total Customers: 7,043
- Churned Customers: 1,869
- Churn Rate: 26.54%
- Average Monthly Charges: ₹4,56,116.60

Visualizations

- Churn Rate by Contract
- Churn Rate by Tenure
- Churn Rate by Payment Method
- Churn Rate by Internet Service
- Average Monthly Charges by Churn
- Churn Rate by Technical Support

The dashboard provides a quick way to compare churn patterns across different customer segments.

Key Insights

The analysis showed that customer churn varies considerably across different customer segments.

Key observations include:

- The overall churn rate is 26.54%.
- Contract type shows noticeable differences in churn behavior.
- Churn varies across different customer tenure groups.
- Payment methods have different churn patterns.
- Internet service type is associated with differences in churn.
- Churned and retained customers show differences in average monthly charges.
- Technical support is another factor worth considering when analyzing customer retention.

These findings can help the business identify high-risk customer groups and develop more targeted retention strategies.

Conclusion

This project demonstrates a complete data analytics workflow, from data cleaning and preprocessing to SQL analysis and dashboard development.

By combining Python, SQL, and Tableau, the project transforms raw customer data into meaningful business insights that can support customer retention and decision-making.

